

INSTRUCTIONS

Please read the instructions carefully before doing the questions.

- You can use materials in your computer, notebook and text book.
- You are **NOT allowed** to use any device to share data with others.

Beside the above conditions, students must follow the following requirements:

1. The work must complete by using Visual Studio 2022++

2. The Framework must be .NET 8.0

3. THIS PART IS VERY IMPORTANT, PLEASE READ IT CAREFULLY AND FOLLOW THE INSTRUCTIONS.

- You are given a database script (.sql file) in Zip file. Execute the script before doing questions.
- **You must use the given solution.**
- **You are not allowed to add any more libraries via NuGet Package Manager into given solution.**

- Submission Guideline:

Submit your work for each question separately. For each question, please:

- Publish your project using the command:

```
dotnet publish -c Release -o ./[QuestionNumber_StudentAccount]
```

Example:

```
dotnet publish -c Release -o ./Q1_trungnthe123432
```

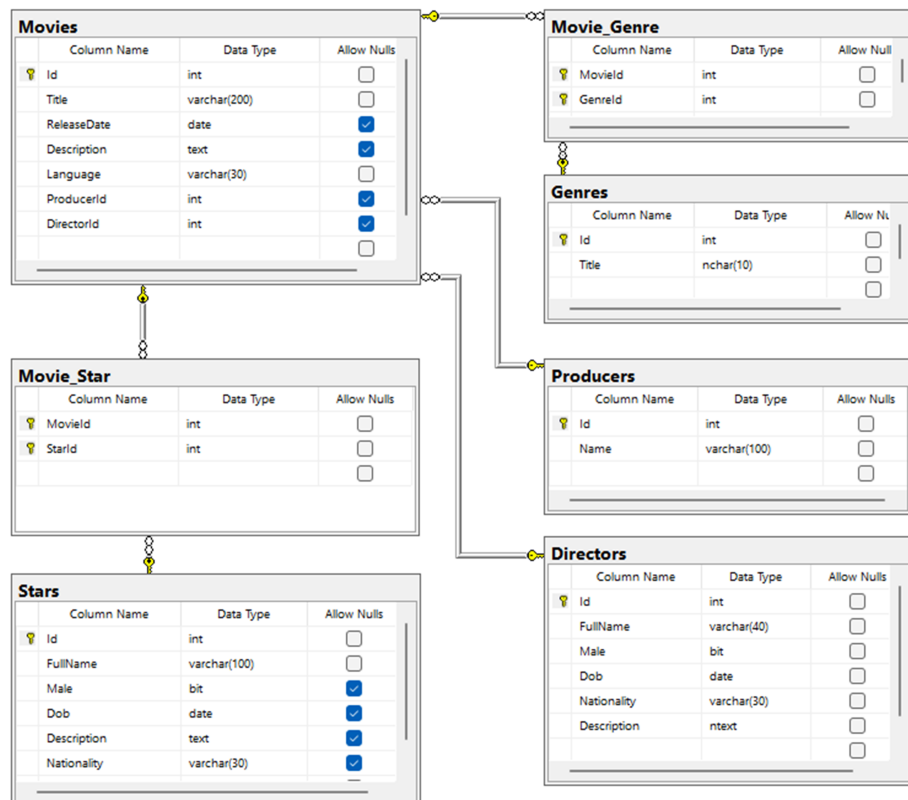
- Submit the root folder of the project into the PEA_Client application.
- If the root folder of the project is too large, you may delete the following subfolders to reduce its size before submitting: /bin, /obj

Just one of above requirements is violated, your work will be considered as invalid.

Question 1:

Question 1.

Use the following database diagram for doing this exam.



In this question, you are asked to write an application that provide some APIs.

Note that you must:

- using the root path **http://localhost:5000**
- using database connection string in the file **appsettings.json** as pattern:

```
{
  "ConnectionStrings": {
    "MyCnn": ""
  }
}
```

You are asked to build three APIs, as following:

1.1

The API at url **/api/director/getdirectors/{nationality}/{gender}**:

- Receives two strings **{nationality}** and **{gender}**.
- using **GET** method.
- Returns all directors of given nationality and gender. The information of each director is the same as in the question part 1.1.

- *dobString* return with format **M/d/yyyy**

Request URL	
<code>http://localhost:5000/api/Director/GetDirectors/usa/male</code>	
Server response	
Code	Details
200	Response body <pre>[{ "id": 1, "fullName": "David Gordon Green", "gender": "Male", "dob": "1975-04-09T00:00:00", "dobString": "4/9/1975", "nationality": "USA", "description": "David Gordon Green was born on April 9, 1975 in Little Rock, Arkansas and George Washington (2000)."</pre> Response headers <pre>content-type: application/json; charset=utf-8 date: Tue, 01 Nov 2022 02:21:44 GMT</pre>

Figure 1 – The result of API at `/api/director/getdirectors/usa/male` (using method GET)

1.2

The API at url `/api/director/getdirector/{id}`:

- Receives an integer `{id}`.
- using GET method.
- Returns all informations (including a list of movies that this person directed) about the director has id of `{id}`.
- *dobString* return with format **M/d/yyyy**
- *genres* and *stars* return **empty list**

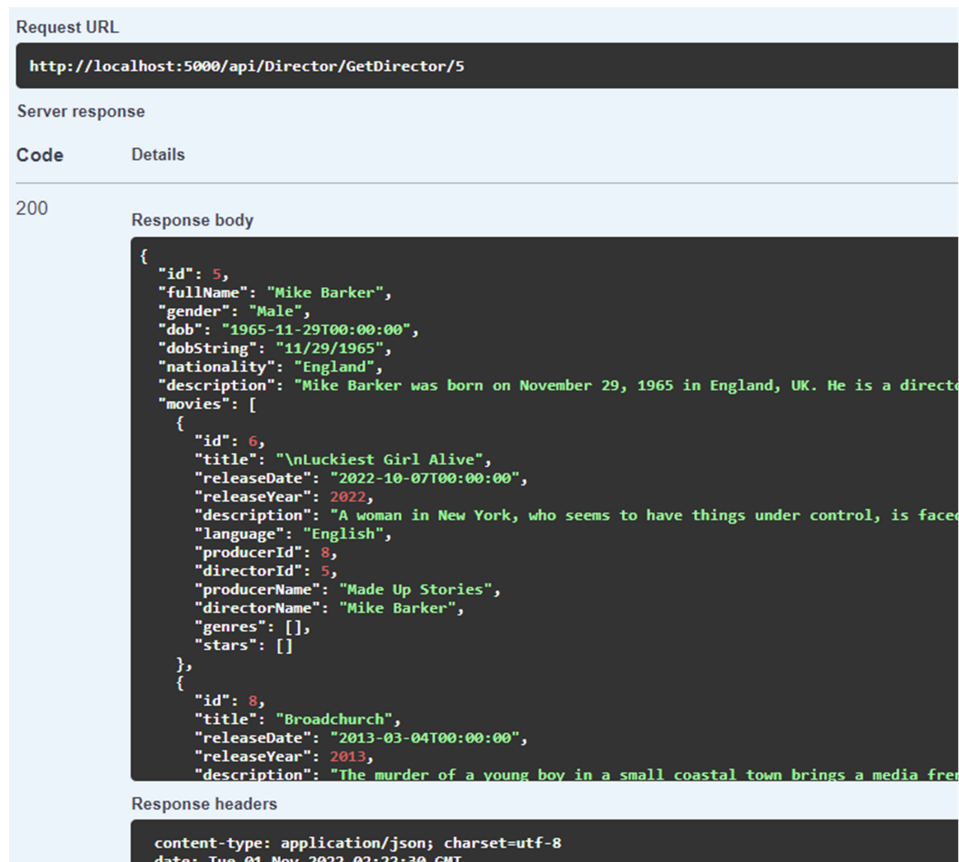


Figure 2 - The result of API at `/api/director/getdirector/5` (using method GET)

1.3

The API at url `/api/director/create` that allows adding a new director to the database:

- using POST method.
- This API insert the new director into the database and returns:
 - The number of records that have been added to the database.
 - Error: Conflict with message **"There is an error while adding."** if any exception occurs during data creation.

See Figure 4, 5, 6 for more detail.

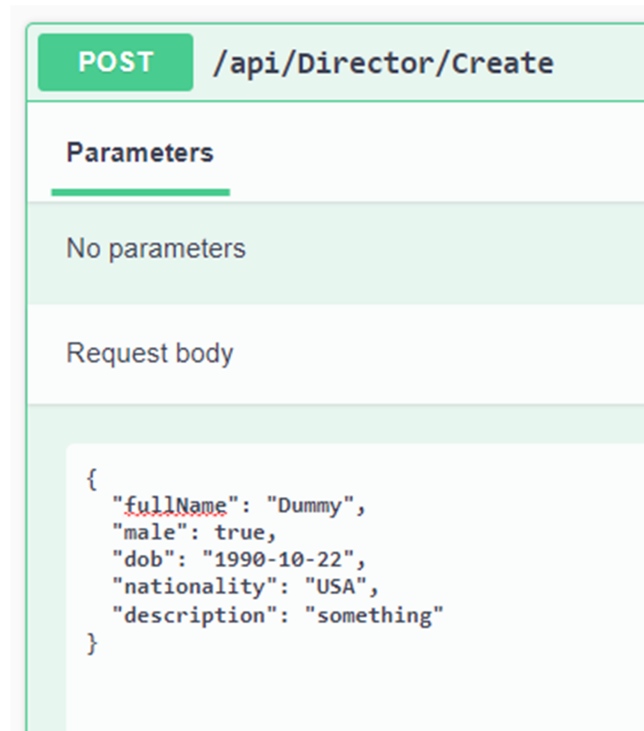


Figure 3 – The input of api at `/api/director/create`

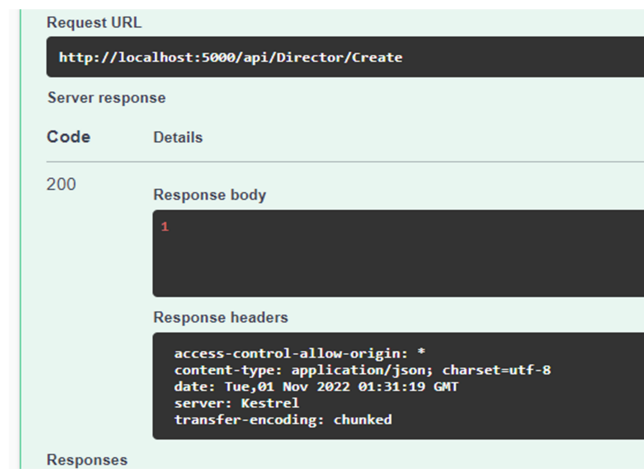


Figure 4 – The result of api at `/api/director/create` in case of success.

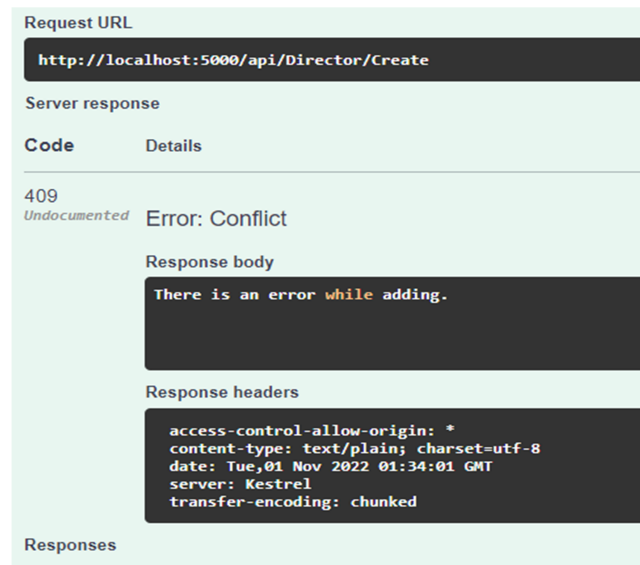


Figure 5 – The result of api at `/api/director/create` when the database server is in stopped state.

Question 2:

Question 2.

In this question, you are asked to write MVC/ Razor Pages model that shows information about students. The application will fetch data by calling pre-existing RESTful APIs hosted at **GivenAPIBaseUrl**. These APIs are provided in a separate project named **GivenAPIs**, which students must run locally to start the API server.

1. Given APIs include:

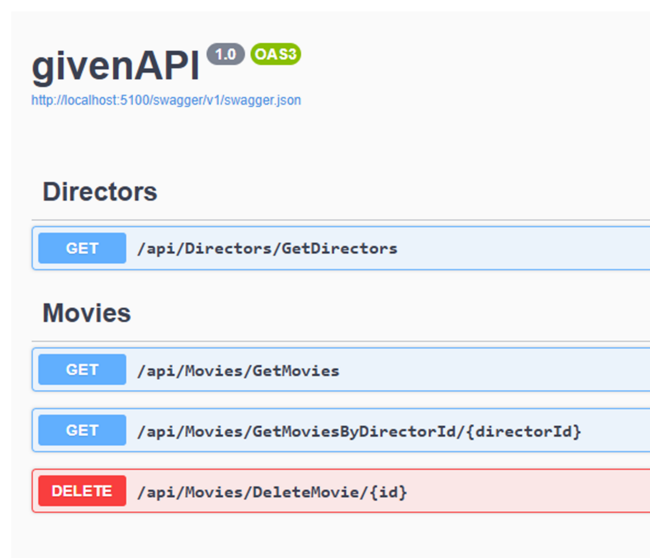


Figure 1 - The given APIs

API	Note
api/Directors/GetDirectors	Return all directors
api/Movies/GetMovies	Return all movies
api/Movies/GetMoviesByDirectorId/{directorId}	Return movies of director
api/Movies/DeleteMovie/{id}	Delete Movie by Id

You can see GivenAPI project for more details.

2. Note (IMPORTANT):

- Students **MUST** use **HttpClient** to make calls to the API.
- The value of **GivenAPIBaseUrl** must be written in **appsettings.json** as following format:

```
{
  "GivenAPIBaseUrl": "http://localhost:5100"
}
```

Students get the **GivenAPIBaseUrl** value from **appsettings.json**, combine it with the **endpoint** to call the API.

- All input and output elements in the HTML source must have an 'id' attribute to ensure accessibility and traceability, student can refer snippet code be provided in **sample.html** in given materials.

Hint:

3. Requirements

Directors

[David Gordon Green](#)
[Aaron Horvath](#)
[David Bruckner](#)
[Mike Barker](#)
[Joseph Kosinski](#)

List of Movies

Title	Release Date	Description	Language	Director	Stars	Action
Halloween Ends	10/14/2022	The saga of Michael Myers and Laurie Strode comes to a spine-chilling climax in this final installment of this trilogy.	English	David Gordon Green	Jamie Lee Curtis, Andi Matichak, James Jude Courtney	Delete
The Super Mario Bros. Movie	04/07/2023	A plumber named Mario travels through an underground labyrinth with his brother, Luigi, trying to save a captured princess. Feature film adaptation of the popular video game.	English	Aaron Horvath	Chris Pratt, Anya Taylor-Joy, Charlie Day	Delete
Hellraiser	10/07/2022	A take on Clive Barker's 1987 horror classic where a young woman struggling with addiction comes into possession of an ancient puzzle box, unaware that its purpose is to summon the Cenobites.	English	David Bruckner	Odessa A'zion, Jamie Clayton	Delete
Luckiest Girl Alive	10/07/2022	A woman in New York, who seems to have things under control, is faced with a trauma that makes her life unravel.	English	Mike Barker		Delete
Broadchurch	03/04/2013	The murder of a young boy in a small coastal town brings a media frenzy, which threatens to tear the community apart.	English	Mike Barker		Delete
Top Gun: Maverick	05/27/2022	After thirty years, Maverick is still pushing the envelope as a top naval aviator, but must confront ghosts of his past when he leads TOP GUN's elite graduates on a mission that demands the ultimate sacrifice from those chosen to fly it.	English	Joseph Kosinski		Delete
The Wolf of Wall Street	01/11/2014	Based on the true story of Jordan Belfort, from his rise to a wealthy stock-broker living the high life to his fall involving crime, corruption and the federal government.	English	Mike Barker	Leonardo DiCaprio, Jonah Hill	Delete

Figure 3 - Required page

The web application has page at url **/Movies/Director_Movie** and is designed (shown in Figure 3), which includes three main parts.

3.1. Display List of Movies

When the first-time access to **/Movies/Director_Movie**

Display all information about Movies in the database in table format. Each row in the table should include the following fields:

- Title
- Release Date (MM/dd/yyyy)
- Description
- Language
- Director (from Directors)
- Stars (from Star) : Each StarName is separated by a "," and there are **no spaces**
- Action Delete

In there:

Each `<td>` tag will have an id in the form **td_{columnName}_{movielfd}** (where movielfd is the id of movie in database, and {columnName} write with camel case format, for example `td_title_0`, `td_releaseDate_0,...`). (see in Figure 4)

Each `<a>` tag [Delete](#) has an id with form **btn_delete_{movielfd}** and inner text is **Delete**

List of Movies

Title	Release Date	Description	Language	Director	Stars	Action
Halloween Ends	10/14/2022	The saga of Michael Myers and Laurie Strode comes to a spine-chilling climax in this final installment of this trilogy.	English	David Gordon Green	Jamie Lee Curtis, Andi Matichak, James Jude Courtney	Delete
The Super Mario Bros. Movie	04/07/2023	A plumber named Mario travels through an underground labyrinth with his brother, Luigi, trying to save a captured princess. Feature film adaptation of the popular video game.	English	Aaron Horvath	Chris Pratt, Anya Taylor-Joy, Charlie Day	Delete
Hellraiser	10/07/2022	A take on Clive Barker's 1987 horror classic where a young woman struggling with addiction comes into possession of an ancient puzzle box, unaware that its purpose is to summon the Cenobites.	English	David Bruckner	Odessa A'zion, Jamie Clayton	Delete
Luckiest Girl Alive	10/07/2022	A woman in New York, who seems to have things under control, is faced with a trauma that makes her life unravel.	English	Mike Barker		Delete
Broadchurch	03/04/2013	The murder of a young boy in a small coastal town brings a media frenzy, which threatens to tear the community apart.	English	Mike Barker		Delete
Top Gun: Maverick	05/27/2022	After thirty years, Maverick is still pushing the envelope as a top naval aviator, but must confront ghosts of his past when he leads TOP GUN's elite graduates on a mission that demands the ultimate sacrifice from those chosen to fly it.	English	Joseph Kosinski		Delete
The Wolf of Wall Street	01/11/2014	Based on the true story of Jordan Belfort, from his rise to a wealthy stock-broker living the high life to his fall involving crime, corruption and the federal government.	English	Mike Barker	Leonardo DiCaprio, Jonah Hill	Delete

Figure 4

3.2. Display all Directors and Filter by Directors

Design list of Directors on the **left** of screen in list `<a>` tag, with

id = **di_{directorId}**, and name of Director inside (see Figure 5, with David Gordon Green has directorId = 1 and Aaron Horvath has directorId = 2)

Directors

[David Gordon Green](#)

di_1

[Aaron Horvath](#)

di_2

[David Bruckner](#)

[Mike Barker](#)

[Joseph Kosinski](#)

Figure 5

When the user click on the link of each director, the list of movies will be updated to include only movies made by selected director (see Figure 6, case click to director Aaron Horvat

Directors

[David Gordon Green](#)
[Aaron Horvath](#)
[David Bruckner](#)
[Mike Barker](#)
[Joseph Kosinski](#)

List of Movies

Title	Release Date	Description	Language	Director	Stars	Action
The Super Mario Bros. Movie	04/07/2023	A plumber named Mario travels through an underground labyrinth with his brother, Luigi, trying to save a captured princess. Feature film adaptation of the popular video game.	English	Aaron Horvath	Chris Pratt, Anya Taylor-Joy, Charlie Day	Delete

Figure 6

3.3. Delete Movie

When users click on [Delete](#), delete the corresponding Movie in the database and go back to [/Movies/Director_Movie](#), make sure the interface is also updated according to the deleted Movie

4. HTML Elements ID

The HTML id requirements are summarized in the table below.

Element	Element Tag	Id
Each cell in the table	<td>	td_{columnName}_{movieId}
Button Delete	<a>	btn_delete_{movieId}
Link to Filter Movies of Director	<a>	di_{directorId}

